# MARIAM OUDEH

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## SUMMARY

Computer Engineering senior with experience spanning embedded hardware development and enterprise IT operations. Background includes building and debugging circuit boards and firmware, along with hands-on exposure to cybersecurity and business application services, policy and procedure documentation, and business process analysis. Strong analytical and problem-solving skills, with experience gathering requirements, documenting processes, and communicating technical work clearly across technical and non-technical stakeholders.

## EDUCATION

**B.S. Computer Engineering | California State University, Sacramento** **Expected Dec 2026**

**A.S. Computer Science | American River College** **2024**

## EXPERIENCE

**Information Technology Student Assistant | FI$Cal, Sacramento, CA** **Aug 2026–Present**

- Rotate across two IT subdivisions, the Enterprise Security Services Office (cybersecurity) and the Business Application Services Office, gaining exposure to how different areas of an IT organization operate and interact.
- Review and update internal policy and procedure documents to align with a newly established template, using SharePoint and Microsoft Office to keep formatting and content consistent and accurate.
- Log and track requests through the office's ticketing system to support documentation and process work.
- Building a working understanding of business process mapping and how individual procedures connect to larger organizational and system-level operations.

**Outreach Specialist / Mentor | TRIO ETS, American River College** **Nov 2022–Jul 2026**

- Maintain and audit a student database of roughly 1,500 records, generating federal compliance reports on tight deadlines.
- Coordinate field trips and STEM workshops for first-generation middle and high school students, managing logistics, permissions, and vendor communication end to end.
- Produce technical documentation, presentations, and reports for internal and federal review.

## RELEVANT PROJECTS

**Senior Capstone: Automated Hydroponic Greenhouse (STM32H753ZIT6)** **Spring–Fall 2026**

- Designed the circuit board layout for the project's control, motor, power, boards using KiCad 10, connecting the microcontroller to all the sensors, displays, and other parts it needs to talk to.
- Lead the lighting system for the project: wrote the firmware that controls brightness across 7 LED channels using PWM and MOSFETs, with different light settings for different plants.
- Found and fixed communication problems between boards using UART and SPI, the two methods the boards use to send data back and forth to each other.
- Designed and ordered a circuit board for the lighting system in EasyEDA, then had it manufactured through JLCPCB.

**Wearable Glucose Monitoring Bracelet (Personal Project)** **Feb 2026–Present**

- Building a wearable on an ESP32-S3 that listens in on BLE signals broadcast by a Dexcom G7 glucose sensor, using the ESP32's built-in Bluetooth radio to pick up the readings.

**Bluetooth-Controlled RC Car (STM32)** **2025**

- Wired the car's circuit (battery pack, motor driver, Bluetooth UART Friend module, STM32 board) and wrote firmware so Bluetooth commands received over UART drive the motors.

**Skystar E-Commerce Site** **2026**

- Built and deployed a full e-commerce website for a small business, including product listings and site design, with a custom admin interface so the non-technical owner can independently add and update products and events.

## SKILLS

Languages/Firmware: C, C++, Python, Java, Verilog, Embedded C/HAL

Protocols/Interfaces: UART, SPI, BLE, PWM, USB CDC

Hardware/Tools: STM32 (STM32H7, STM32L4), ESP32-S3, KiCad, EasyEDA, oscilloscope, logic analyzer, schematic capture, PCB layout, lab equipment/OS setup (Windows/Linux)

Other: Technical documentation, data analysis, cross-team communication, SharePoint, Microsoft Office, ticketing systems

## ACTIVITIES

Girls Who Code | Society of Women Engineers | Forensic Engineering Club | Muslim Student Association